Andy Chen 陳奕翔

Taipei City, Taiwan

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EXPERIENCE

ARDGE
Jan. 2026 – Present

Software Engineer
Taipei, Taiwan

- **vLLM Scheduler & GPU Kernel Optimization:** Implemented custom algorithm in both vLLM scheduler and GPU kernel on GB300 Desktop, achieving **2.4x** tok/s at single request and **4.3x** tok/s at 32 concurrent requests.
- **KV Cache Management & Compression:** Evaluated and benchmarked **PagedAttention**, **RadixAttention**, **LMCache**, and **TurboQuant** for memory efficiency.
- **Enterprise OCR System:** Engineered a high-throughput production OCR application achieving **95.8%+ accuracy** at a processing speed of **1 page/sec**.
- **Large-Scale RAG System (20M+ Chunks):** Designed and optimized a custom vector search algorithm, reducing query retrieval latency to **under 500ms** across twenty-million-scale document repositories.
- **Customer Service Agent Harness:** Architected an agentic customer support framework centered on user experience, leveraging **LLM tool-calling capabilities** to streamline complex client-facing workflows.

National Center for High-performance Computing (NCHC)
July 2025 – Dec. 2025

Software Engineer Intern
Hsinchu City, Taiwan

- **LLM Application & RAG Architecture:** Led the development of an intelligent customer service system. Implemented a robust RAG workflow via **LangChain** utilizing by-page and by-file chunking strategies. Integrated a **BM25 hybrid search** alongside **PostgreSQL (pgvector)** to maximize document retrieval accuracy.
- **Microservices & Scalability:** Packaged applications into **Docker** images and deployed them in a **Kubernetes** environment. Verified through stress testing that the system stably supports over **4,000 concurrent users**.

PROJECTS

LensCopy — Full-Stack AI-Powered OCR SaaS Platform
2025 – Present

- **Independently SaaS:** Designed, built, and deployed a complete document-to-text extraction platform with a **Vue 3** frontend, **FastAPI** backend, and a multi-stage ML pipeline — serving **6,000+ unique visitors/month** with **10,000+ pages processed**, on a freemium subscription model with geo-aware pricing across **37+ countries**.
- **Multi-GPU ML Pipeline:** Engineered a high throughput pipeline architecture where a **Deep-learning layout detector** overlaps with a **Vision-language model** inference across **multiple GPUs** via **vLLM**, utilizing continuous batching and **asyncio** for **200+ concurrent** inference requests with automatic VRAM-aware batch calibration.
- **Production Infrastructure:** Orchestrated **10+ Docker Compose services** across 2 GPUs — vLLM, pipeline, FastAPI, Vue/Nginx proxy, PostgreSQL, Redis, **SeaweedFS** (S3 storage), admin panel — with full observability via **Prometheus**, **Grafana**, and Dozzle.

Lightweight Code Completion Model Fine-Tuning
Jul. 2025 – Sept. 2025

- **End-to-End Pipeline Design:** Architected comprehensive pipelines for data synthesis, preprocessing, training, and benchmarking (**Human-Eval**) to fine-tune Small Language Models (**Gemma3-270M**, **Llama-3.2-1B**, **Qwen2.5-Coder-1.5B**) utilizing **PyTorch**.
- **Automated Optimization:** Integrated **Optuna** for automated hyperparameter tuning, systematically discovering optimal configurations that significantly enhanced convergence rate and coding capabilities.

- **HPC Orchestration:** Accelerated intensive data processing and training pipelines across **NVIDIA H200** GPU clusters; authored robust **Slurm** batch scripts to manage job scheduling, resource allocation, and distributed workloads.
- **Publication:** First-author paper, *"Enhancing Small Language Models in Agentic Pipelines with Prompt Engineering and Fine-Tuning,"* accepted at the Taiwan Academic Network Conference (**TANET 2025**).

OPEN SOURCE

Taiwan Document OCR (zh-TW) — Open-Source OCR Dataset Collection 2025 – Present

- **Dataset Creation & Curation:** Created and published **4 block-level OCR datasets** (**1.2M+ annotated blocks**, **5,000+ downloads**) on **Hugging Face**, covering Traditional Chinese government and financial documents from Taiwan spanning **1928–2024**: financial statements, Legislative Yuan gazettes, Judicial Yuan publications, and urban planning records.
- **Structured Block-Level Annotations:** Each record pairs a document-image crop with block-level **OCR text, bounding box, and layout type** (paragraph, table, title, seal, etc.) with document/page/block indexing, supporting downstream OCR, layout-analysis, and document-understanding model training and evaluation.

RT-DETR-Model-Benchmark — Multi-Framework Inference Benchmark 2026

- **Multi-Backend Benchmarking:** Benchmarked the **PP-DocLayoutV3** layout-detection model across three inference backends — **HuggingFace Transformers/PyTorch**, **ONNX Runtime**, and native **PaddlePaddle** — under matched conditions (GPU sync, identical pre/post-processing) to compare latency, throughput, and RAM/GPU memory.
- **Bug Discovery & Fix:** Identified and fixed a **preprocessing normalization bug** in the community ONNX conversion that was silently dropping detections; published reproducible benchmark scripts and full methodology notes.

EDUCATION

National Taipei University of Technology (NTUT) Aug. 2022 – June 2026

Bachelor of Science in Computer Science and Information Engineering Taipei, Taiwan

- **Research Focus**: Advanced AI applications, LLM fine-tuning, Pre-training data collection/cleaning pipelines, and Retrieval-Augmented Generation (RAG).

Taipei Municipal Daan Vocational High School Aug. 2019 – June 2022

High School Diploma in Electronic Engineering Taipei, Taiwan

SKILLS

Programming & OS : Python, JavaScript/TypeScript, C/C++, SQL, Linux/Ubuntu, Bash/Shell Scripting

AI/ML & Frameworks : Large Language Models (LLMs), LangChain, vLLM, PyTorch, OpenCV, scikit-learn, Hugging Face Transformers, RAG, Optuna, BM25 Hybrid Search, Vector Databases (pgvector)

Web & Frontend : Vue 3, Pinia, Tailwind CSS, Vite, Vue I18n, HTML/CSS

Tools & Infrastructure : Docker, Kubernetes, Nginx, GitLab CI/CD, GitHub Actions, Prometheus, Grafana, FastAPI, PostgreSQL, Redis, SeaweedFS, Git, Slurm

Languages : English (TOEIC 865, CEFR B2), Mandarin (Native)